

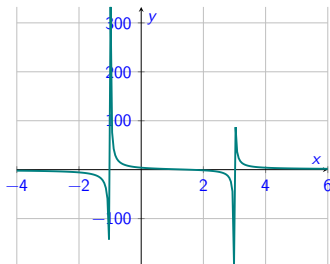
2627-MA140: Week 01, Lecture 3 (L03)

# Polynomials and Rational Functions

**Dr Niall Madden**

University of Galway

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# Today we rationalize the following topics:

- 1 News!
  - Tutorials
  - Tutorial sheet
- 2 Polynomials
  - Quadratic
  - Sketching polynomials
- 3 Rational Functions
  - Long division
- 4 Towards Partial Fractions
  - Quadratic factorization
- 5 Exercises

See also Sections 1.2 and 7.4(!)  
of [https://math.libretexts.org/Bookshelves/Calculus/Calculus\\_\(OpenStax\)/01%3A\\_Functions\\_and\\_Graphs](https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)/01%3A_Functions_and_Graphs)

Slides are on canvas, and at  
<https://www.niallmadden.ie/2627-MA140/>



Tutorials start **next** week. The schedule is:

| Teams | Time           | Venue            | Leader |
|-------|----------------|------------------|--------|
| 1–2   | Tuesday 15:00  | ENG- <b>2003</b> | ST     |
| 3–4   | Tuesday 15:00  | ENG- <b>2034</b> | DC     |
| 11–12 | Thursday 11:00 | ENG- <b>2002</b> | ST     |
| 9–11  | Thursday 11:00 | ENG- <b>3035</b> | DC     |
| 5–6   | Friday 13:00   | Eng- <b>2002</b> | ST     |
| 7–8   | Friday 13:00   | Eng- <b>2035</b> | DC     |

*Do these data agree with information you have been provided?*

Would you be interested to taking a tutorial through Irish? (Show of hands?)

You don't have to complete a graded assignment next week. However, this is a “practice” one available. See <https://universityofgalway.instructure.com/courses/58477/assignments/162051>

During tutorials, the tutor will solve some similar questions. You can access the **tutorial sheet** at the link above (and on Canvas: [Modules... Tutorial Sheets](#)).

The Tutorial Sheet has questions that are nearly identical to your own version.

# Polynomials

Yesterday, we saw that...

## Polynomials

**Polynomials** are functions of the form

$$y = f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x^1 + a_0, \quad x \in \mathbb{R},$$

where  $a_0, a_1, \dots, a_n$  are real numbers called the **coefficients** of the polynomial. The number  $n$  is called the **degree** of the polynomial.

**Examples:**

**Example: quadratic**

$x^2 - 2x - 3$  is a **quadratic** polynomial: it has degree  $n = 2$ .

There are many occasions when we want to **factorise** such quadratics, meaning we write them as the product of a pair of linear polynomials.

For example, we can **factorise**  $x^2 - 2x - 3$  as

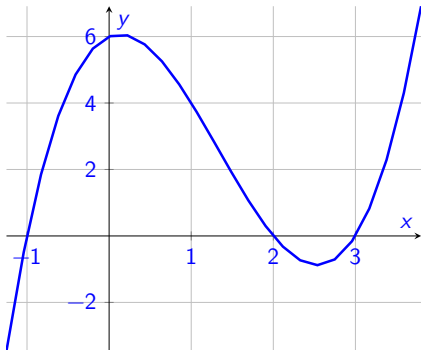
$$x^2 - 2x - 3 = (x - 3)(x + 1)$$

It is important to note that not all quadratic polynomials can be factorised as two linear polynomials. Such quadratics are called **irreducible**.

For example,  $x^2 + 1$  is irreducible.

**Example**

$y = x^3 - 4x^2 + x + 6$  is a **cubic** function with degree  $n = 3$ .



**Fact**

A polynomial function of degree  $n$  has **up to**  $n-1$  turning points (“bends”).

**Examples:**

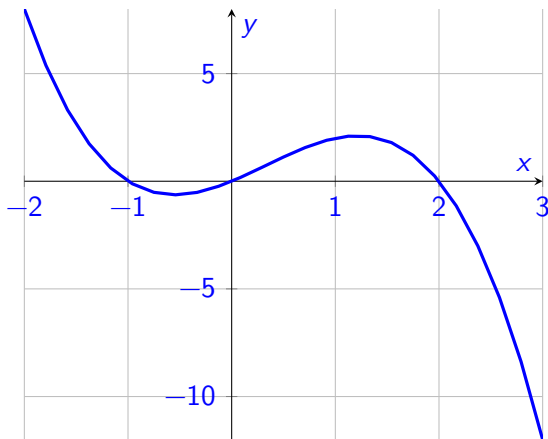
When sketching the graph of a function, we first find the **intercepts**:

- ▶ The **y-intercept** is where the graph of the function cuts the  $y$ -axis: found by letting  $x = 0$ .
- ▶ The **x-intercepts** are where the function's graph cuts the  $x$ -axis. These points are also called the **roots** (or **zeros**). To find them, set  $y$  equal to zero and solve for  $x$ .

### Example

Sketch the graph of  $y = -x^3 + x^2 + 2x$

Actual plot of  $y = -x^3 + x^2 + 2x$



## Rational Functions

**Rational Functions** have the general form

$$f(x) = \frac{p(x)}{q(x)},$$

where  $p(x)$  and  $q(x)$  are polynomials.

- ▶ If degree of  $p(x) < \text{degree of } q(x)$ ,  
 $f(x)$  is called a **strictly proper rational function**.
- ▶ If degree of  $p(x) = \text{degree of } q(x)$ ,  
 $f(x)$  is called a **proper rational function**.
- ▶ If degree of  $p(x) > \text{degree of } q(x)$ ,  
 $f(x)$  is called an **improper rational function**.

# Rational Functions

An improper or proper rational function can always be expressed as a polynomial plus a strictly proper rational function, for example by algebraic division.

## Example

$$\frac{4x^3 + 4x^2 + 4}{x^2 - 3} = 4x + 4 + \frac{12x + 16}{x^2 - 3}$$

For the previous example, we can work this out ourselves using **Long Division** to divide numerator by denominator:

**Example**

Use long division to show that

$$\frac{3x^4 + 2x^3 - 5x^2 + 6x - 7}{x^2 - 2x + 3} = 3x^2 + 8x + 2 - \frac{14x + 13}{x^2 - 2x + 3}$$

Try this yourself!

# Towards Partial Fractions

**Rational Functions** have the general form  $f(x) = \frac{p(x)}{q(x)}$ , where  $p(x)$  and  $q(x)$  are polynomials.

An (proper) rational function can often be written as a sum of simpler ones: **partial fractions**.

For example

$$\frac{8x - 12}{x^2 - 2x - 3}$$

can be written as

$$\frac{3}{x - 3} + \frac{5}{x + 1}$$

**Check:** (*next slide*)

# Towards Partial Fractions

## Towards Partial Fractions

In that last example, we manipulated  $\frac{3}{x-3} + \frac{5}{x+1}$  to show that it is the **partial fraction** representation of  $\frac{8x-12}{x^2-2x-3}$

In the next section, we'll work on the problem of *finding* the **partial fraction** representation of a rational function.

A key step is *factorization of polynomials*.

In the previous example, a key step was noting that  $x^2 - 2x - 3 = (x - 3)(x + 1)$ . That is, “ $x^2 - 2x - 3$  is factorized as  $(x - 3)(x + 1)$ ”

Often, we'll want to factorize quadratics of the form

$$x^2 + bx + c,$$

meaning writing it as a product of two linear factors:

$$x^2 + bx + c = (x + q)(x + s).$$

To factorize, we need to find  $q$  and  $s$  such that

$$q + s = b \quad qs = c.$$

For example,

$$x^2 + 5x + 6 = (x + q)(x + r)$$

then  $q + s = 5$  and  $qs = 6$ .

By inspection, we can see that we can take  $q = 2$  and  $s = 3$ :

$$x^2 + 5x + 6 = (x + 2)(x + 3).$$

Factoring is useful because it makes it easy to find the roots:

$$(x + 2)(x + 3) = 0$$

implies

$$x = -2 \quad \text{or} \quad x = -3.$$

## Exercise 1.3.1

Sketch the graphs of

(i)  $y = 5x^2 - 7$

(ii)  $y = x^2 - 4x + 3$

(iii)  $y = x^3 - 6x^2 - 11x - 6$